Práctica Dos: Sistema cardiovascular

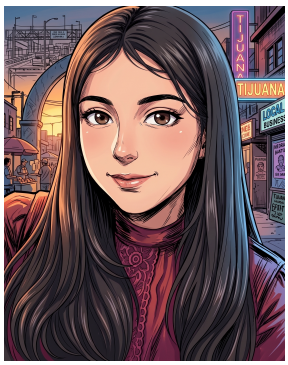
Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

Tecnológico Nacional de México [TecNM - Tijuana], Blvd. Alberto Limón Padilla s/n, C.P. 22454, Tijuana, B.C., México

Table of Contents

Información general....1
Datos de la simulación....1
Respuesta Hipotenso, Normotenso y Hipertenso....2
Respuesta Normotenso....2
Respuesta Hipertenso....3
Modelo de EDOs....3
Funcion: Respuesta de la simulacion....4

Información general



Nombre del alumno: **Nayeli Bonilla Martinez**

Número de control: **22212381**

Correo institucional: **l22212381@tectijuana.edu.mx**

Asignatura: **Modelado de Sistemas Fisiológicos**

Docente: **Dr. Paul Antonio Valle Trujillo; paul.valle@tectijuana.edu.mx**

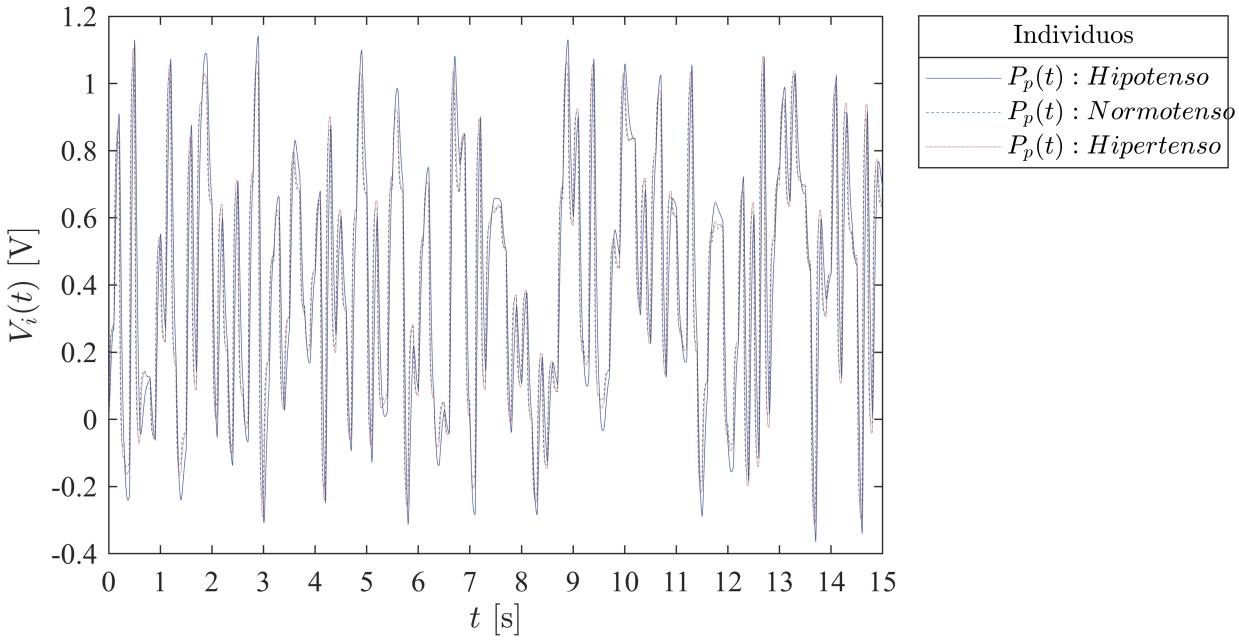
Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system ('Sistemab')
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-2';
```

```
x = sim('Sistemab', parameters);
t_data = (0:0.1:15)';
t = (0:0.001:15)';
Pa = interp1(t_data, x.Pa, t,'previous');
writematrix(Pa,'signal.xlsx');
```

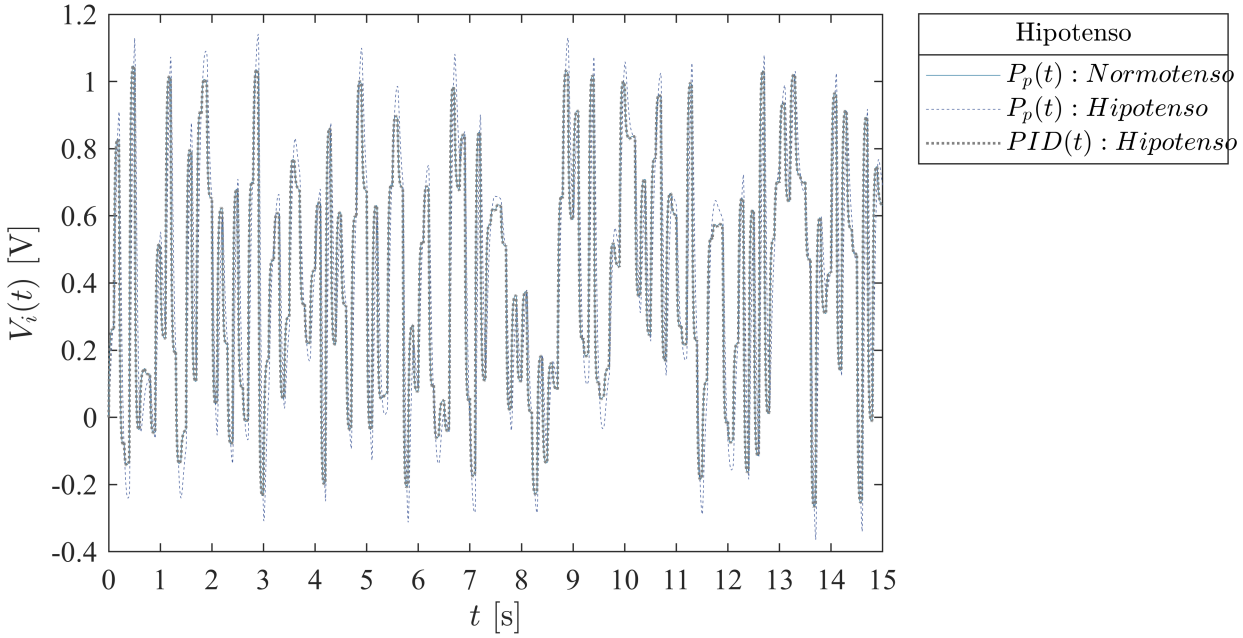
Respuesta Hipotenso, Normotenso y Hipertenso

```
plotsignals(x.t,x.PpL, x.PpN, x.PpH,'Individuos')
```



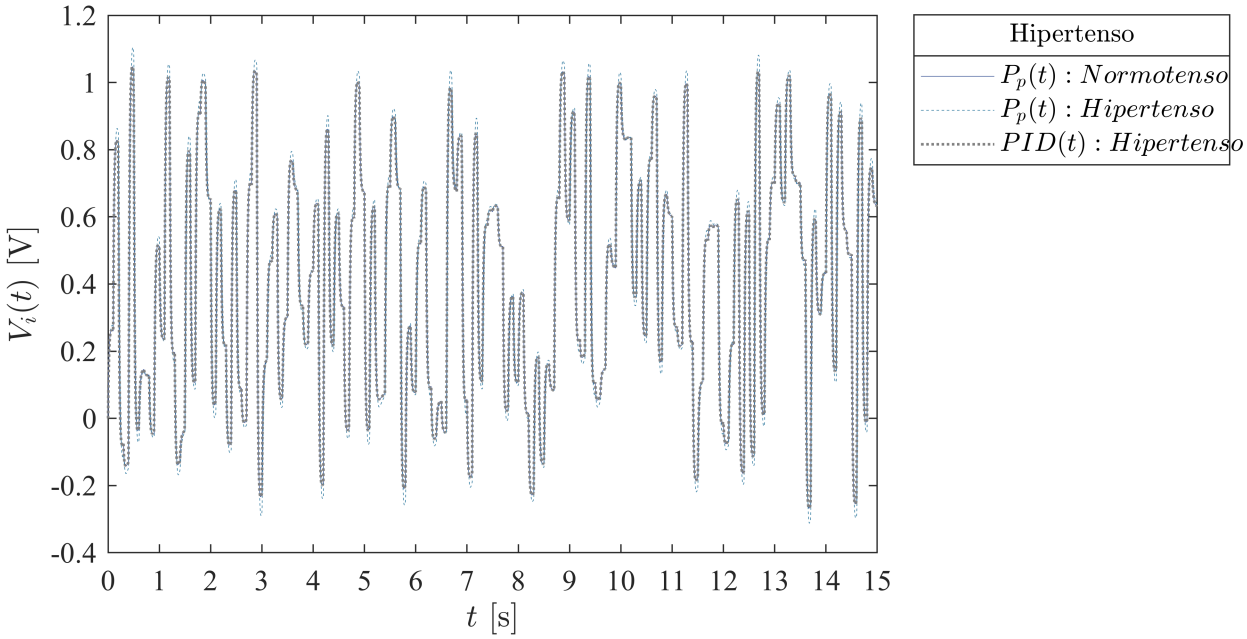
Respuesta Normotenso

```
plotsignals(x.t,x.PpN, x.PpL, x.PIDL,'Hipotenso')
```



Respuesta Hipertenso

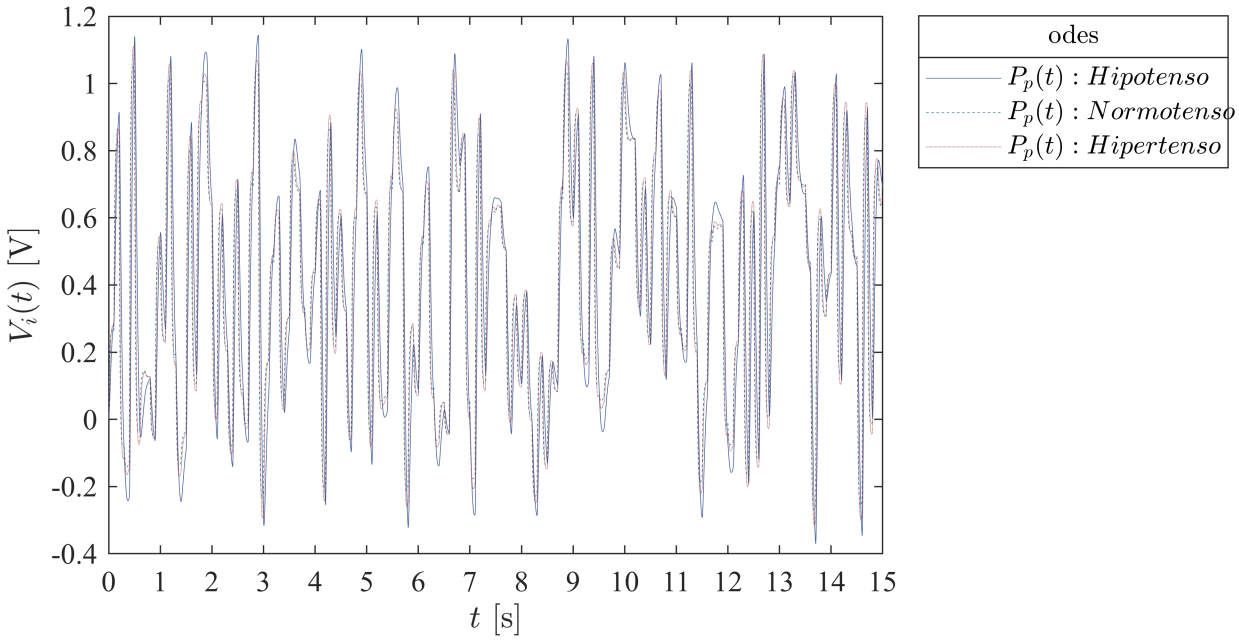
```
plotsignals(x.t,x.PpN, x.PpH, x.PIDH,'Hipertenso')
```



Modelo de EDOs

```
par_hipo = [0.01,0.001,1.00, 3.500];
par_normo = [0.01,0.001,1.00, 1.500];
par_hiper = [0.01,0.001,1.00, 1.000];
PpN = cardio (par_normo,Pa,t);
PpL = cardio (par_hipo,PpN,t);
```

```
PpH = cardio (par_hiper,PpN,t);
plotsignals(t,PpL,PpN, PpH, 'odes')
```




```
function Pp= cardio(par,Pa,t)
  Z=par(1); L= par(2); R= par(3);C= par(4); dt=t(2);
  n= round(t(end)/dt);
  FL= zeros(length(t),1); FL(1)=0;
  Pp= zeros(length(t),1); FL(1)=0;
  for i =1:n
     FL(i+1) = FL(i) + (-Pp(i)/L + Pa(i)/L)*dt;
     Pp(i+1) = Pp(i) + (FL(i)/C - (1/(Z*C) + 1/(R*C)) * Pp(i) + Pa(i)/(Z*C))*dt;
  end

 end
```

Funcion: Respuesta de la simulacion

```
function plotsignals(t,x1,x2,x3,caso)
set(figure(),'color','w')
set(gcf,'units','inches','position',[1,1,7,3.5])
set(gca,'fontname','times new roman','fontsize',11)

hold on; grid off; box on;
clr1 = [66, 91, 154]/255;
clr2 = [48, 122, 160]/255;
clr3 = [151, 40, 40]/255;
clr4 = [0.5, 0.5, 0.5];
```

```
if caso=="Individuos"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr1) %hipotenso
    plot(t,x2,'--','LineWidth',0.25,'Color',clr2)% Normotenso
    plot(t,x3,':','LineWidth',0.25,'Color',clr3)%Hipertenso
L = legend('$P_p(t): Hipotenso$','$P_{p}(t): Normotenso$','$P_p(t): Hipertenso$');
elseif caso=="Hipotenso"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr2) %hipotenso
    plot(t,x2,'--','LineWidth',0.25,'Color',clr1)% Normotenso
    plot(t,x3,':','LineWidth',1,'Color',clr4)% Controlador del Hipotenso
L = legend('$P_p(t): Normotenso$','$P_{p}(t): Hipotenso$','$PID(t): Hipotenso$');
elseif caso=="Hipertenso"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr1) %hipotenso
    plot(t,x2,'--','LineWidth',0.25,'Color',clr2)% Normotenso
    plot(t,x3,':','LineWidth',1,'Color',clr4)%Controlador del Hipertenso
L = legend('$P_p(t): Normotenso$','$P_{p}(t): Hipertenso$','$PID(t): Hipertenso$');
elseif caso=="odes"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr1) %hipotenso
    plot(t,x2,'--','LineWidth',0.25,'Color',clr2)% Normotenso
    plot(t,x3,':','LineWidth',0.25,'Color',clr3)%Hipertenso
L = legend('$P_p(t): Hipotenso$','$P_{p}(t): Normotenso$','$P_p(t): Hipertenso$');
end

set(L,'interpreter','latex','location','northeastoutside','orientation','vertical')
title(L,caso)
xlabel('$t$ [s]','interpreter','latex')
ylabel('$V_i(t)$ [V]','interpreter','latex')
xlim([0 15]); xticks(0:1:15)
ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)


exportgraphics(gcf,[caso,'_matlab.pdf'],'ContentType', 'vector')
end
```